

NQS: A Composite Network Quality Score for 5G/LTE Mobile Networks

Signal Monitor Project

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Abstract

We present the Network Quality Score (NQS), a composite 0–100 metric that aggregates four radio and transport-layer measurements—RSRP, SINR, round-trip latency, and download throughput—into a single, user-perceivable quality grade. The scoring model draws on ETSI TR 103 559 per-KPI normalisation, the IQX exponential sensitivity hypothesis for latency, the Weber-Fechner psychophysical law for throughput perception, and Shannon capacity theory for weighting SINR. When individual metrics are unavailable (e.g., throughput during idle periods), the algorithm dynamically redistributes weights among the remaining inputs. NQS is deployed in a real-time Android monitoring application targeting 5G SA, 5G NSA, and LTE networks, where it drives dashboard gauges, map colour-coding, and historical trend analysis. Early field observations indicate that NQS tracks subjective user experience more faithfully than any single radio metric alone.

1. Introduction

Assessing mobile network quality from the user's perspective is inherently multi-dimensional. A strong signal (high RSRP) does not guarantee fast data transfer if interference is high (low SINR); low latency is irrelevant if throughput is insufficient for the application's needs. Industry approaches such as the Rohde & Schwarz Network Performance Score and Opensignal's experience metrics address this by combining multiple KPIs into a single figure of merit.

This paper describes **NQS** (Network Quality Score), a lightweight composite metric designed for real-time on-device computation on Android. NQS operates on four inputs that are always (or nearly always) available from the Android telephony and networking APIs: Reference Signal Received Power (RSRP), Signal-to-Interference-plus-Noise Ratio (SINR), HTTP round-trip latency, and passive or active download throughput. The design prioritises three goals:

- **Perceptual fidelity:** the score should correlate with what a user actually feels—buffering, lag, dropped connections—rather than with abstract radio measurements.
- **Robustness to missing data:** when throughput is unavailable (phone idle), the score must remain meaningful by redistributing weight to the remaining metrics.
- **Computational simplicity:** the formula must run in real time on a mobile device with negligible CPU and memory overhead.

2. Related Work

2.1 ETSI TR 103 559 — Network Performance Score

The European Telecommunications Standards Institute published TR 103 559 (v1.2.1, October 2023) defining a three-level aggregation framework for mobile network benchmarking. At Level 1, each Key Performance Indicator (KPI) is normalised to a common 0–1000 scale using boundary values that represent the thresholds of human perception (perceptual saturation). Level 2 aggregates per-service

scores (telephony, video streaming, web browsing, data transfer), and Level 3 produces an overall network score weighted by service importance and geographic coverage. NQS adopts the Level 1 normalisation philosophy and the principle of perceptual saturation at the bounds of each metric.

2.2 ITU-T G.107 E-Model

The E-Model, standardised as ITU-T Recommendation G.107, computes an *R-factor* (0–100) for narrowband voice quality. The R-factor accounts for signal-to-noise ratio, codec impairment, delay, and an *advantage factor* (+10 for mobile) that reflects user tolerance for access convenience. The R-factor maps to Mean Opinion Score (MOS, 1–5) via a cubic polynomial. NQS borrows the 0–100 scale and the concept of mapping to categorical labels (Excellent, Good, Fair, Poor, Bad) analogous to the MOS bands.

2.3 IQX Hypothesis

Fiedler, Hossfeld, and Tran-Gia (IEEE Network, 2010) proposed the IQX hypothesis: Quality of Experience (QoE) degrades *exponentially* as Quality of Service (QoS) disturbance increases, with the sensitivity proportional to the current QoE level. Formally: $\text{QoE} = a \cdot \exp(-b \cdot d) + c$, where d is the disturbance magnitude. This means that a small increase in latency from 20 ms to 40 ms is perceived as a larger degradation than the same 20 ms increase from 200 ms to 220 ms. NQS applies this exponential transform to the latency sub-score.

2.4 Weber-Fechner Law

The Weber-Fechner psychophysical law states that human perception of stimulus change is proportional to the logarithm of the stimulus intensity. Applied to throughput: doubling speed from 1 to 2 Mbps is perceived as a much larger improvement than doubling from 50 to 100 Mbps. Reichl et al. (2010) demonstrated this logarithmic relationship holds for web browsing and video streaming QoE. NQS uses a logarithmic normalisation for the throughput sub-score.

2.5 Shannon Capacity

Shannon’s channel capacity theorem establishes that maximum throughput scales as $C = B \cdot \log_2(1 + \text{SINR})$. This logarithmic relationship between SINR and achievable data rate justifies assigning SINR the highest weight among the radio-layer metrics: SINR is the single best predictor of the throughput ceiling the network can deliver.

3. Methodology

3.1 Input Metrics

NQS consumes four metrics, each chosen for both technical significance and direct user-perceivable impact:

Metric	Source	Unit	User Impact
RSRP	CellSignalStrength Lte / Nr	dBm	Coverage: can the user connect at all?
SINR	CellSignalStrength Lte / Nr	dB	Throughput ceiling: how clean is the channel?

Metric	Source	Unit	User Impact
Latency	HTTP HEAD to configurable host	ms	Responsiveness: browsing, gaming, video calls
Throughput	TrafficStats delta or active probe	Mbps	Perceived speed: streaming, downloads

Table 1. Input metrics and their user-perceivable significance.

3.2 Per-Metric Normalisation

Each raw metric is mapped to a sub-score s_i in $[0, 1]$ using a metric-specific transform function. All transforms incorporate **perceptual saturation**: improvements beyond the excellent threshold or degradation below the dead threshold produce no further score change.

RSRP (Linear). The signal level maps linearly between boundary values:

$$s_RSRP = \text{clamp}((RSRP - (-120)) / ((-80) - (-120)), 0, 1)$$

where -120 dBm represents a dead zone (score 0) and -80 dBm represents excellent coverage (score 1). Values outside this range are clamped.

SINR (Linear). Channel quality maps linearly:

$$s_SINR = \text{clamp}((SINR - (-5)) / (20 - (-5)), 0, 1)$$

where -5 dB is an unusable channel and +20 dB is excellent. The linear transform is appropriate here because Shannon capacity is logarithmic in $(1 + SINR)$, meaning the SINR-to-throughput relationship is already compressive.

Latency (Exponential / IQX). Following the IQX hypothesis, latency sub-score decays exponentially:

$$d = \text{clamp}((ms - 20) / (300 - 20), 0, 1)$$

$$s_Latency = \exp(-5.0 * d)$$

The decay constant $\beta = 5.0$ is chosen so that 100 ms latency yields approximately 0.45 (mid-range), reflecting the research finding that latency degradation is perceived exponentially more severely at low latencies. At 20 ms the score is 1.0; at 300 ms it drops to approximately 0.007.

Throughput (Logarithmic / Weber-Fechner). Download speed is normalised on a logarithmic scale:

$$s_DL = (\ln(\text{clamp}(Mbps, 0.5, 50)) - \ln(0.5)) / (\ln(50) - \ln(0.5))$$

This ensures that doubling throughput from 1 to 2 Mbps contributes a larger score gain than doubling from 25 to 50 Mbps, matching psychophysical perception.

3.3 Weighting

Base weights are assigned to reflect each metric's predictive power for user-perceived quality:

Metric	Weight	Rationale
SINR	0.30	Best single predictor of throughput ceiling (Shannon capacity theorem)

Metric	Weight	Rationale
RSRP	0.25	Primary coverage indicator; determines connectivity
Latency	0.25	Directly perceived in interactive applications (gaming, browsing, calls)
Throughput	0.20	Most tangible metric, but often unavailable when phone is idle

Table 2. Base weights and rationale.

3.4 Dynamic Weight Redistribution

When a metric is unavailable (most commonly throughput, which is null when the phone is idle), its weight is redistributed proportionally among the remaining metrics. If the available weights sum to W , each available metric's effective weight becomes w_i / W . This ensures the composite score always sums to 100% of available evidence.

For example, when throughput is unavailable ($W = 0.25 + 0.30 + 0.25 = 0.80$), the effective weights become: RSRP = 0.3125, SINR = 0.375, Latency = 0.3125.

3.5 Composite Score

The final NQS is computed as:

$$\text{NQS} = 100 * \text{SUM}(w_i / W * s_i), \text{ clamped to } [0, 100]$$

where the sum runs over all available metrics, w_i is the base weight, W is the sum of available base weights, and s_i is the normalised sub-score.

4. Score Interpretation

NQS maps to five categorical bands, analogous to the ITU-T MOS scale. The boundaries are chosen to align with the R-factor bands in ITU-T G.107:

NQS Range	Label	Colour	User Experience
80 - 100	Excellent	Green	4K streaming, fast gaming, instant page loads
60 - 79	Good	Light Green	HD streaming, smooth video calls, responsive browsing
40 - 59	Fair	Orange	SD video, basic browsing, occasional buffering or lag
20 - 39	Poor	Dark Orange	Frequent stalls, slow page loads, intermittent connectivity
0 - 19	Bad	Red	Effectively unusable for most data-dependent applications

Table 3. NQS bands and corresponding user experience.

5. Implementation

NQS is implemented as a stateless Kotlin object (**NetworkQualityScore**) in the Signal Monitor Android application. The compute function accepts nullable metric values, gracefully handling missing data via dynamic weight redistribution. Key implementation details:

- **Real-time computation:** NQS is calculated for each sample as it is collected by the foreground monitoring service (every 1–30 seconds, configurable). The computation is pure arithmetic with no allocations, completing in under 1 microsecond.
- **Dashboard integration:** A circular gauge displays the live NQS with a trend indicator (improving/degrading) based on the recent sample window.
- **Map visualisation:** Map circles and heatmap tiles are coloured by NQS rather than raw RSRP, providing a holistic view that accounts for interference and latency conditions not visible from signal strength alone.
- **Historical analysis:** The statistics screen reports average NQS over configurable time ranges (24h, 7d, 30d), enabling trend monitoring.
- **Dead zone detection:** Areas are flagged as dead zones when the average NQS-contributing RSRP falls below -100 dBm across 3 or more samples in a ~100 m grid cell.

6. Limitations and Future Work

The current NQS model has several known limitations that suggest avenues for future improvement:

- **Fixed weights:** The base weights are derived from domain expertise and literature review rather than empirical fitting to user satisfaction data. A large-scale MOS study correlating NQS with subjective ratings would enable data-driven weight optimisation via regression or AHP.
- **No jitter or packet loss:** Android does not expose jitter or packet loss metrics through public APIs. These are critical for real-time communication quality (VoIP, video conferencing). Future versions could estimate jitter from sequential latency measurements.
- **Service-agnostic:** NQS produces a single score regardless of the application in use. A user streaming 4K video has different quality requirements than one sending text messages. Per-service weighting profiles (as in ETSI TR 103 559 Level 2) would improve fidelity.
- **Temporal stability:** The current score is instantaneous. Recent research (arXiv:2510.21162, 2025) suggests that networks with the same average metrics but different stability patterns are perceived differently. Incorporating a variance penalty could address this.
- **Machine learning:** The explicit formula could be replaced or augmented by a trained model (random forest, neural network) once sufficient labelled QoE data is collected through the app's deployment.

7. Conclusion

NQS provides a principled, lightweight composite metric for mobile network quality that combines academic rigour with practical deployability. By grounding each sub-score transform in established psychophysical and telecommunications research (ETSI normalisation, IQX exponential sensitivity, Weber-Fechner logarithmic perception, Shannon capacity theory), the score faithfully reflects user-perceived quality while remaining computationally trivial. The dynamic weight redistribution mechanism ensures meaningful scores even when individual metrics are temporarily unavailable. Deployed in the Signal Monitor Android application, NQS serves as the primary quality indicator across dashboard, map, and statistical views.

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